Supplementary Material for Epoch-Aware Threshold Estimation for Blink Detection in Simulated-Driving EEG

Rodney Petrus
Balandong, Nur Syahirah Roslan, Syaimaa' Solehah Mohd Radzi, Muhamad Hafiz Abd Latif, and Tong Boon Tang

I. Electrode Montages and Regions

Table and figure numbers in this document carry the prefix S. The 32 channels supplied to the pipeline for Cao2018 were Fp1, F3, Fp2, F4, FT7, FC3, C3, FC4, FT8, C4, TP7, CP3, TP8, CP4, T5, P3, O1, T6, P4, O2, F7, T3, Fz, FCz, Cz, Pz, Oz, F8, T4, CPz, A1, and A2. For the Internal corpus the 32 channels were Fp1, AF3, F3, Fp2, AF4, F4, FC5, FC1, C3, FC2, FC6, C4, CP5, CP1, CP2, CP6, P8, P3, PO3, O1, PO4, P4, P7, O2, F7, T7, Fz, Cz, Pz, Oz, F8, and T8.

The electrodes were divided into four anatomical regions (frontal, central, parietal, and occipital) and each region into left and right hemispheres, as in [1], as shown in Fig. S1. The resulting eight regions (frontal left, frontal right, central left, central right, parietal left, parietal right, occipital left, and occipital right) are abbreviated FL, FR, CL, CR, PL, PR, OL, and OR.

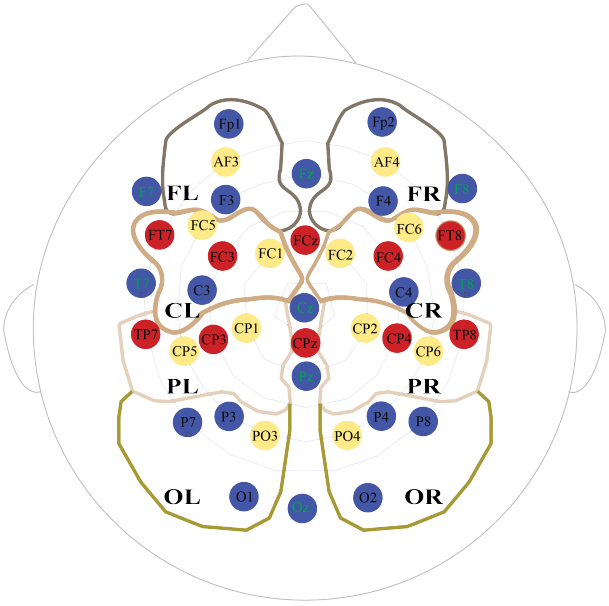


Fig. S1. Scalp locations of the 32 electrodes used in this study, divided into eight anatomical regions (frontal left/right, central left/right, parietal left/right, and occipital left/right, abbreviated FL, FR, CL, CR, PL, PR, OL, and OR). Blue circles mark electrodes present in both the Internal and Cao2018 montages, yellow circles mark electrodes present only in the Internal montage, and red circles mark electrodes present only in the Cao2018 montage.

II. Per-Electrode Figures

Fig. S2 shows the per-electrode session-mean F_1 when all 32 channels were supplied. On Internal, C3 reached 61.06% whereas C4 reached only 1.85%, while on Cao2018 C3 and C4 performed similarly at 31.08% and 30.18%, so the C3–C4 asymmetry was specific to Internal. Figs. S3 and S4 show the regional reruns and their comparison with the full montage, and Fig. S5 shows the single-electrode runs.



Fig. S2. Per-electrode detection F_1 for every electrode in the four curated anatomical regions, scored in the full-montage run, coloured by scalp region. Each bar is one electrode's session-mean F_1 under the complete pipeline using all 32 channels.

III. Per-Electrode Statistics for the Regional and Single-Electrode Analyses

Tables S1 and S2 report, for every electrode, the session-level F_1 obtained when the pipeline was rerun with only the electrodes of that electrode's region supplied, alongside the F_1 of the same electrode in the full-montage run. Tables S3 and S4 report the corresponding comparison for the single-electrode runs, in which each electrode was supplied alone.



Fig. S3. Per-electrode detection F_1 in the regional rerun of each electrode's own region, coloured by region. Each bar is one electrode's session-mean F_1 when only the electrodes of its region were supplied to the pipeline.

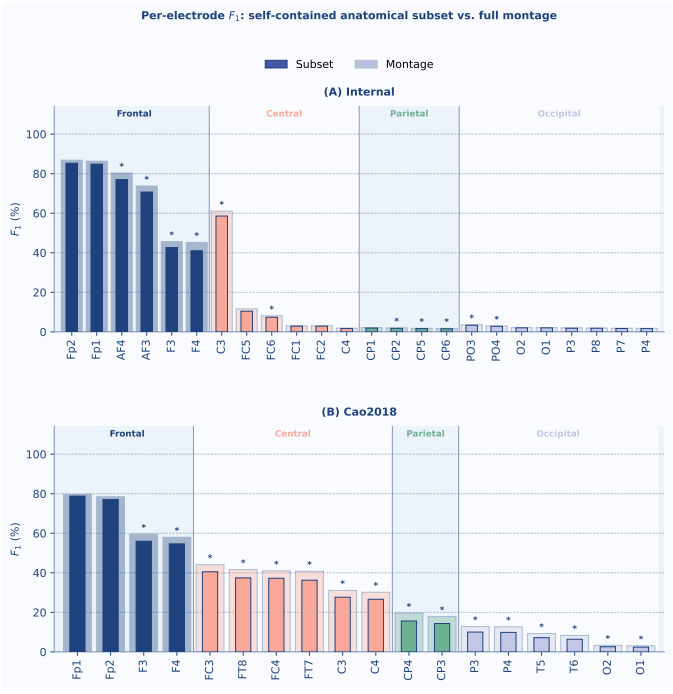


Fig. S4. Per-electrode comparison of F_1 for each electrode in the rerun of its own anatomical region (narrow, opaque bars) and in the full-montage run (wide, translucent bars), for (A) Internal and (B) Cao2018. Bars show session-mean F_1. Electrodes are grouped by the anatomical regions in Fig. S1 and ordered by descending regional-rerun F_1 within each region. Comparisons are paired by session using two-tailed Wilcoxon signed-rank tests, with p-values Bonferroni-corrected separately over the electrodes tested within each dataset. An asterisk marks adjusted $p < 0.05$.

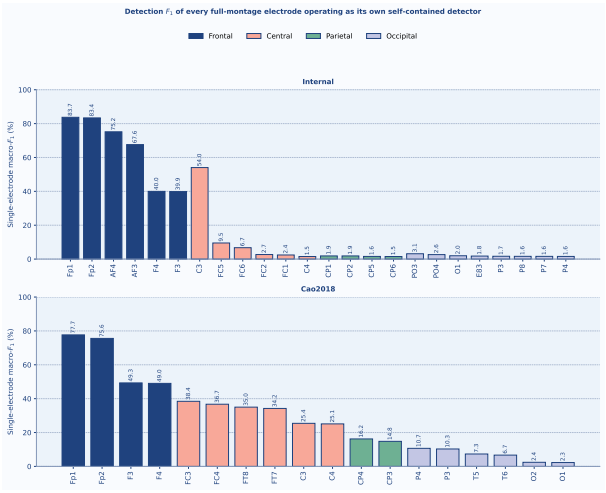


Fig. S5. Detection F_1 of every electrode in the four curated anatomical regions operating as its own self-contained detector, in which the complete pipeline was rerun on that one electrode alone, coloured by scalp region.

TABLE S1

PER-ELECTRODE F_1 IN THE REGIONAL RERUN VERSUS THE FULL-MONTAGE RUN ON THE INTERNAL DATASET. EACH ELECTRODE'S SESSION-LEVEL F_1 IN THE REGIONAL RERUN, IN WHICH ONLY THE ELECTRODES OF THAT ELECTRODE'S REGION WERE SUPPLIED TO THE PIPELINE, WAS PAIRED WITH ITS SESSION-LEVEL F_1 IN THE FULL-MONTAGE RUN (ALL 32 CHANNELS SUPPLIED) ON THE SESSIONS COMMON TO BOTH RUNS AND COMPARED WITH A TWO-TAILED WILCOXON SIGNED-RANK TEST PAIRED BY SESSION. AN ASTERISK ($*$) MARKS ELECTRODES WHOSE DIFFERENCE WAS SIGNIFICANT AT THE 0.05 LEVEL AFTER BONFERRONI CORRECTION OVER THE 24 ELECTRODES TESTED WITHIN THE INTERNAL DATASET. ΔF_1 IS THE REDUCED-CONFIGURATION F_1 MINUS THE FULL-MONTAGE F_1, IN PERCENTAGE POINTS, SO NEGATIVE VALUES INDICATE LOWER F_1 IN THE REDUCED CONFIGURATION. WITHIN EACH REGION, ELECTRODES ARE ORDERED BY DESCENDING REGIONAL-RERUN F_1.

Region	Electrode	F_1 regional rerun (%)	F_1 full montage (%)	ΔF_1	Sig.
Frontal	Fp2	85.28	86.82	−1.54	
	Fp1	84.84	86.32	−1.48	
	AF4	77.04	80.32	−3.28	*
	AF3	70.70	73.76	−3.06	*
	F3	42.68	45.65	−2.96	*
	F4	41.00	45.23	−4.24	*
Central	C3	58.59	61.06	−2.47	*
	FC5	10.45	11.60	−1.14	
	FC6	7.40	8.10	−0.71	*
	FC1	2.89	3.07	−0.18	
	FC2	2.89	3.11	−0.22	
	C4	1.73	1.85	−0.12	
Parietal	CP1	1.92	2.19	−0.27	
	CP2	1.78	2.02	−0.24	*
	CP5	1.64	1.80	−0.16	*
	CP6	1.54	1.75	−0.20	*
Occipital	PO3	3.35	3.60	−0.25	*
	PO4	2.81	2.98	−0.17	*
	O2	2.05	2.13	−0.09	
	O1	2.04	2.13	−0.09	
	P3	1.83	2.05	−0.22	
	P8	1.82	1.94	−0.12	
	P7	1.71	1.82	−0.11	
	P4	1.65	1.77	−0.12	

TABLE S2

PER-ELECTRODE $\boldsymbol{F_1}$ IN THE REGIONAL RERUN VERSUS THE FULL-MONTAGE RUN ON THE CAO2018 DATASET. EACH ELECTRODE'S SESSION-LEVEL $\boldsymbol{F_1}$ IN THE REGIONAL RERUN, IN WHICH ONLY THE ELECTRODES OF THAT ELECTRODE'S REGION WERE SUPPLIED TO THE PIPELINE, WAS PAIRED WITH ITS SESSION-LEVEL $\boldsymbol{F_1}$ IN THE FULL-MONTAGE RUN (ALL 32 CHANNELS SUPPLIED) ON THE SESSIONS COMMON TO BOTH RUNS AND COMPARED WITH A TWO-TAILED WILCOXON SIGNED-RANK TEST PAIRED BY SESSION. AN ASTERISK ($*$) MARKS ELECTRODES WHOSE DIFFERENCE WAS SIGNIFICANT AT THE 0.05 LEVEL AFTER BONFERRONI CORRECTION OVER THE 18 ELECTRODES TESTED WITHIN THE CAO2018 DATASET. $\boldsymbol{\Delta F_1}$ IS THE REDUCED-CONFIGURATION $\boldsymbol{F_1}$ MINUS THE FULL-MONTAGE $\boldsymbol{F_1}$, IN PERCENTAGE POINTS, SO NEGATIVE VALUES INDICATE LOWER $\boldsymbol{F_1}$ IN THE REDUCED CONFIGURATION. WITHIN EACH REGION, ELECTRODES ARE ORDERED BY DESCENDING REGIONAL-RERUN $\boldsymbol{F_1}$.

Region	Electrode	F_1 regional rerun (%)	F_1 full montage (%)	ΔF_1	Sig.
Frontal	Fp1	78.80	79.68	−0.88	
	Fp2	77.05	78.45	−1.40	
	F3	55.97	59.51	−3.55	$*$
	F4	54.64	57.99	−3.35	$*$
Central	FC3	40.52	44.13	−3.61	$*$
	FT8	37.44	41.63	−4.19	$*$
	FC4	37.25	40.99	−3.73	$*$
	FT7	36.26	40.76	−4.49	$*$
	C3	27.67	31.08	−3.42	$*$
	C4	26.62	30.18	−3.56	$*$
Parietal	CP4	15.63	19.56	−3.93	$*$
	CP3	14.37	17.84	−3.46	$*$
Occipital	P3	10.04	12.80	−2.77	$*$
	P4	9.86	12.67	−2.81	$*$
	T5	7.20	9.26	−2.06	$*$
	T6	6.39	8.30	−1.91	$*$
	O2	2.61	3.33	−0.72	$*$
	O1	2.40	3.10	−0.70	$*$

TABLE S3

PER-ELECTRODE $\boldsymbol{F_1}$ IN THE SINGLE-ELECTRODE RUN VERSUS THE FULL-MONTAGE RUN ON THE INTERNAL DATASET. EACH ELECTRODE'S SESSION-LEVEL $\boldsymbol{F_1}$ IN THE SINGLE-ELECTRODE RUN, IN WHICH THAT ELECTRODE ALONE WAS SUPPLIED TO THE PIPELINE, WAS PAIRED WITH ITS SESSION-LEVEL $\boldsymbol{F_1}$ IN THE FULL-MONTAGE RUN (ALL 32 CHANNELS SUPPLIED) ON THE SESSIONS COMMON TO BOTH RUNS AND COMPARED WITH A TWO-TAILED WILCOXON SIGNED-RANK TEST PAIRED BY SESSION. AN ASTERISK ($*$) MARKS ELECTRODES WHOSE DIFFERENCE WAS SIGNIFICANT AT THE 0.05 LEVEL AFTER BONFERRONI CORRECTION OVER THE 24 ELECTRODES TESTED WITHIN THE INTERNAL DATASET. $\boldsymbol{\Delta F_1}$ IS THE REDUCED-CONFIGURATION $\boldsymbol{F_1}$ MINUS THE FULL-MONTAGE $\boldsymbol{F_1}$, IN PERCENTAGE POINTS, SO NEGATIVE VALUES INDICATE LOWER $\boldsymbol{F_1}$ IN THE REDUCED CONFIGURATION. WITHIN EACH REGION, ELECTRODES ARE ORDERED BY DESCENDING SINGLE-ELECTRODE-RUN $\boldsymbol{F_1}$.

Region	Electrode	F_1 single-electrode (%)	F_1 full montage (%)	ΔF_1	Sig.
Frontal	Fp1	83.73	86.32	−2.59	
	Fp2	83.37	86.82	−3.45	
	AF4	75.16	80.32	−5.16	$*$
	AF3	67.59	73.76	−6.17	$*$
	F4	39.98	45.23	−5.26	$*$
	F3	39.85	45.65	−5.79	$*$
Central	C3	54.05	61.06	−7.02	$*$
	FC5	9.51	11.60	−2.09	$*$
	FC6	6.73	8.10	−1.37	
	FC2	2.69	3.11	−0.42	
	FC1	2.42	3.07	−0.65	
	C4	1.51	1.85	−0.34	$*$
Parietal	CP1	1.86	2.19	−0.32	$*$
	CP2	1.86	2.02	−0.15	
	CP5	1.57	1.80	−0.23	
	CP6	1.53	1.75	−0.22	
Occipital	PO3	3.15	3.60	−0.45	$*$
	PO4	2.65	2.98	−0.34	$*$
	O1	1.97	2.13	−0.16	
	O2	1.78	2.13	−0.36	$*$
	P3	1.66	2.05	−0.39	$*$
	P8	1.62	1.94	−0.32	$*$
	P7	1.61	1.82	−0.21	$*$
	P4	1.57	1.77	−0.20	

TABLE S4

PER-ELECTRODE $\boldsymbol{F_1}$ IN THE SINGLE-ELECTRODE RUN VERSUS THE FULL-MONTAGE RUN ON THE CAO2018 DATASET. EACH ELECTRODE'S SESSION-LEVEL $\boldsymbol{F_1}$ IN THE SINGLE-ELECTRODE RUN, IN WHICH THAT ELECTRODE ALONE WAS SUPPLIED TO THE PIPELINE, WAS PAIRED WITH ITS SESSION-LEVEL $\boldsymbol{F_1}$ IN THE FULL-MONTAGE RUN (ALL 32 CHANNELS SUPPLIED) ON THE SESSIONS COMMON TO BOTH RUNS AND COMPARED WITH A TWO-TAILED WILCOXON SIGNED-RANK TEST PAIRED BY SESSION. AN ASTERISK ($*$) MARKS ELECTRODES WHOSE DIFFERENCE WAS SIGNIFICANT AT THE 0.05 LEVEL AFTER BONFERRONI CORRECTION OVER THE 18 ELECTRODES TESTED WITHIN THE CAO2018 DATASET. $\boldsymbol{\Delta F_1}$ IS THE REDUCED-CONFIGURATION $\boldsymbol{F_1}$ MINUS THE FULL-MONTAGE $\boldsymbol{F_1}$, IN PERCENTAGE POINTS, SO NEGATIVE VALUES INDICATE LOWER $\boldsymbol{F_1}$ IN THE REDUCED CONFIGURATION. WITHIN EACH REGION, ELECTRODES ARE ORDERED BY DESCENDING SINGLE-ELECTRODE-RUN $\boldsymbol{F_1}$.

Region	Electrode	F_1 single-electrode (%)	F_1 full montage (%)	ΔF_1	Sig.
Frontal	Fp1	77.65	79.68	−2.03	
	Fp2	75.59	78.45	−2.86	
	F3	49.30	59.51	−10.21	$*$
	F4	49.03	57.99	−8.96	$*$
Central	FC3	38.44	44.13	−5.70	$*$
	FC4	36.72	40.99	−4.27	$*$
	FT8	34.99	41.63	−6.63	$*$
	FT7	34.22	40.76	−6.54	$*$
	C3	25.43	31.08	−5.66	$*$
	C4	25.10	30.18	−5.08	$*$
Parietal	CP4	16.22	19.56	−3.34	$*$
	CP3	14.80	17.84	−3.04	$*$
Occipital	P4	10.66	12.67	−2.01	$*$
	P3	10.28	12.80	−2.52	$*$
	T5	7.32	9.26	−1.94	$*$
	T6	6.66	8.30	−1.64	$*$
	O2	2.42	3.33	−0.91	$*$
	O1	2.28	3.10	−0.82	$*$

REFERENCES

[1] B. E. Butler and L. J. Trainor, "Sequencing the cortical processing of pitch-evoking stimuli using eeg analysis and source estimation," *Frontiers in Psychology*, vol. Volume 3 - 2012, 2012. [Online]. Available: https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2012.00180